



Interactive gaze and finger controlled HUD for cars

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Abstract

Modern infotainment systems in automobiles facilitate driving at the cost of secondary tasks in addition to the primary task of driving. These secondary tasks have considerable chance to distract a driver from his primary driving task, thereby reducing safety or increasing cognitive workload. This paper presents an intelligent interactive head up display (HUD) on the windscreen of the driver that does not require them to take eyes off from road while undertaking secondary tasks like playing music, operating vent controls, watching navigation map and so on. The interactive HUD allows interaction in the form of pointing and selection just like traditional graphical user interfaces, however tracking operators' eye gaze or finger movements. Additionally, the system can also estimate drivers' cognitive load and distraction level. User studies show the system improves driving performance in terms of mean deviation from lane in an ISO 26022 lane changing task compared to touchscreen system and participants can undertake ISO 9241 pointing tasks in less than 2 s on average inside a Toyota Etios car.

Keywords Automotive UI · Multimodal interaction · Eye gaze controlled system · Virtual touch · Head up display

1 Introduction

Consumer electronics market has seen a plethora of interactive devices aimed to enhance user experience or facilitate human–machine interaction. In automotive domain, the need for improvement of user interfaces is increasing as the complexity of the secondary tasks increases day by day with integration of new technologies. In automotive environment, existing infotainment systems use electronic displays on central stack or head up displays as output modalities while input is taken through touchscreen, physical buttons and voice recognition technologies. To facilitate human machine interaction, researchers have already explored use of haptic feedback [12], hand gesture tracking based input [31], personalizing instrument displays and predictive models to help drivers in regular driving tasks like parking [17, 30]. Researchers attempted to eliminate or reduce visual search using gesture recognition techniques but most systems either require to remember a set of gestures (AirGesture System

[26]) or relative positions of screen items (BullsEye system [40]). Additionally, such systems worked inferior to a touchscreen system in terms of driving performance or secondary task. It is interesting to note that most systems can not completely exclude visual displays and visual feedback. Visual search pattern is also reported to be distinctive for different drivers [22]. This invention has explored the possibility to leverage the visual search and eye gaze fixations as a mean of controlling electronic displays inside a car.

Recent advancement in infra-red based eye gaze trackers has significantly increased research and industrial use of gaze tracking technology. Eye tracking is the process of measuring either the point of gaze (where one is looking) or the motion of an eye relative to the head. An eye tracker is a device for measuring eye positions and eye movement. Research on developing eye tracker investigates reducing the cost of existing infra-red based trackers as well as increasing their accuracy. Most commercially available eye gaze trackers used infra-red cameras to track the eye gaze of users. However accurately tracking eye gaze in different conditions of ambient light is a challenging problem. The most commonly used technique is pupil centre and corneal reflection technique. The eye gaze tracker has inbuilt Infrared Light Emitting Diode (LED) that illuminates the eye and infrared-sensitive video takes rapid

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pictures of eye. The LED reflects small amount of light off the cornea & through the pupil onto the retina. The bright pupil allows image processor to locate centre of pupil. The eye gaze tracker can then locate where the person is looking on the screen based on the relative positions of the pupil centre and corneal reflection within the video image of the eye. A calibration procedure requires to follow (means staring at) a shape around screen or a designated position in space tunes parameters for each individual user with respect to the two-dimensional screen or three dimensional space. However there also exist eye gaze trackers that utilise cameras in visible spectrum like webcam or high speed video cameras although those systems are either less accurate (for webcam) or costlier (for high speed video cameras) than infrared trackers.

Although research involving analysis of eye gaze dates back to early nineteenth century, until recently eye gaze trackers were mainly used for analysing ocular parameters for reading and a variety of human machine interaction (HMI) tasks. With progress in processor speed and image processing algorithms, it has also become possible to use the gaze tracking technology in real time to control a screen pointer in a direct manipulation interface. Gaze controlled interfaces have already been investigated and used for assistive technology [8], automotive and aviation environment. Kern [21] and Poitschke [33] have already reported user studies involving simulated driving tasks while comparing eye gaze controlled interface with traditional touchscreen system. An eye gaze controlled HUD

- can be placed even out of the reach of the driver,
- will relieve drivers from taking their hands off from steering wheel or gear handle which may reduce response times and
- will be helpful for elderly driver who has reduced range of motion at shoulder due to age related physical impairment like arthritis.

In the next section, we presented a brief literature survey on use of eye gaze tracking technology in automotive environment. Sections 3 and 4 identified problems with existing state-of-the-art and described the interactive HUD in details. Section 5 presented different issues in terms of developing content for the HUD and described our approach in terms of designing user interface elements for the same. We described five user studies in Sect. 6 on various components of the interactive HUD involving both driving simulator and real car followed by concluding remarks at Sect. 7.

2 Related work

Research on eye gaze tracking started even in late eighteenth century although controlling an electronic interface using eye gaze or finger movement is relatively new concept.

For eye gaze controlled interface, Zhai [42] pioneered in MAGIC system, which aimed to use eye gaze track directly to improve pointing, in particular homing on a target using mouse. Ashdown et al. [2] addressed the issue of head movement while tracking eye-gaze in a multiple monitor scenario. They used head tracking to switch pointer across screens, which was preferred by participants, but in effect increased pointing time. Dostal et al. [13] addressed similar issues by detecting which monitor the user is looking at through analysing webcam video. The Sideways system [43] even eliminates personalized calibration and can scroll contents of a display screen by detecting eye gaze. The system identifies whether users are looking at the middle or sides of a display and if they are looking to the sides, the system scrolls content at the middle. Both Dostal's system and Sideways system does not use precise x and y coordinates to a move a mouse pointer. Fu and Huang [18] proposed an input system hMouse, which moves a pointer based on head movement. They detected head movement by analysing video input and their system is found to outperform another similar system called CameraMouse [10]. Fejtova's [16] Magic Key system also uses a webcam like CameraMouse but the pointer is moved in screen based on position of nose (nosetrills to be precise). Selection is done by eye blinks. Bates [4] multimodal eye tracking system allows zooming portion of screen using a polhemus tracker. Zander et al. [41] combine a BCI system with eye-gaze tracking, where EEG generated by imagining a rinsing action is trained to make a selection. However, their system had limited success in reducing pointing times. Lutteroth [23] and colleagues investigated different dwell time durations for selecting target for an eye gaze controlled interface, although their study involved only selecting target at the middle of the screen. Later they extended the study [23] for selecting hyperlinks in a webpage but dwell time based selection would be inappropriate in automotive environment as it requires operators staring away from road for selecting target. Pfeuffer and Voelker explored fusing touchscreen interaction with gaze controlled system by using eye gaze for object selection and touch interaction for object manipulation. Pfeuffer [32] explored desktop computing tasks like image searching and map navigation while Voelker [38] investigated multi-screen display, which is more advanced in terms of coordinate mapping between horizontal and vertical displays compared to Dostal's [13] system. However, our proposed work uses eye gaze for not only object selection but also for cursor movement.

Gaze controlled interfaces were already patented for automotive environment. For example, Mondragon et al. [28] presented an eye gaze controlled smart display for passengers of vehicles. Users may point and select icons on the display by staring at appropriate portion of the screen. However, our proposed system is intended to be used by drivers and selection of target has to be faster than dwelling or

staring away from road. Seder et al. [34] presented a graphic projection display for drivers showing objects in road ahead. Users can select object on the projection display using different input modalities including eye gaze. However, the patent does not address a method of improving accuracy of the gaze tracking itself and it does not intend to operate the dashboard as the proposed system. Poitschke's [33] study compared gaze controlled dashboard with touchscreen and reported higher reaction time for gaze controlled dashboard. They did not propose any intelligent algorithm to reduce pointing and selection times for gaze controlled interface.

Existing patents on eye gaze tracking mainly concerned about developing eye gaze tracking hardware [39] and using it to plot users' gaze location in a two-dimensional screen [27]. There are a set of patents which proposes to modify part of display based on eye gaze of users [14, 19] or modifying rendering of a web browser [25]. Farrell [15] proposed to expand targets and proportionally contract other part of display based on eye gaze track of users although their system used a separate pointing device for cursor control instead of eye gaze. Vahtola [37] and Jang [20] proposed eye gaze enabled touch screen system where eye gaze tracking is used to confirm the touch input. The eye gaze tracking system required another cursor tracking program to operate. Pfeuffer and Voelker explored fusing touchscreen interaction with gaze controlled system by using eye gaze for object selection and touch interaction for object manipulation. Pfeuffer [32] explored desktop computing tasks like image searching and map navigation while Voelker [38] investigated multi-screen display. Our proposed system does not require another cursor tracking program or a touch sensitive display and both eye gaze and virtual touch tracker can operate on their own. The eye tracker requires a physical switch press to confirm selection which only gives binary input to the system instead of a full cursor tracking program as required in Jang [20] or Vahtola [37]'s systems. Since the proposed system is operable without a touchscreen, the system can be deployed as a Head Up Display outside the reach of drivers' hands. Additionally, a set of intelligent algorithm in the proposed system significantly reduced pointing and selection times for gaze controlled interface. A patent is filed in the Indian Patent Office with application number 201941009219.

3 Problem statement

Drivers need to take their eyes off the road and physically touch the dashboard to operate the infotainment system. Nowadays drivers often put their iPad or smartphone on top of the dashboard to avoid looking down but the device itself occlude part of the windscreen. The voice output system is not often useful for different language speakers and in noisy environment. The proposed system shows information in

front of driver and they can operate it either just by looking at the display or small finger movements. The key problems are

1. Developing an eye gaze and finger controlled interface that enables pointing and selection tasks as fast as existing touch screen system but for a Head Up Display (HUD)
2. Developing a HUD in automotive environment that changes its content like a direct manipulation interface following drivers' eye gaze and finger movement
3. Placement of the HUD, eye gaze and finger tracking sensors inside a car that does not interfere with safe operation of other driving instruments like steering wheel, gear box and so on.
4. Developing appropriate content for the interactive HUD system that covers common secondary tasks
5. Evaluating performance of the system for a person in driving position
6. Evaluating performance of the system in a moving vehicle under variable vibrating and lighting conditions.

4 Proposed approach

The proposed system consists of hardware and software part described below while Fig. 1 below shows a picture of the set up inside a car. The system can be deployed in two ways

1. Using reflection from a screen with high brightness on a semi-transparent sheet placed on the windscreen (Fig. 1a)
2. Overlaying controls on the video feed of a tablet computer (Fig. 1b)

The next section describes the hardware set up used to arrange the HUD in the reflected screen set up while the following sections describe details of the gaze and finger-controlled software interfaces and content of the system.

4.1 Holder set up

The hardware part of the invention will be placed on the dashboard in front of the driver. It should be designed in such a way so as not to distract or block the operator's field of view. It should display the labels and functions clearly and the operator should be able to interact with minimum effort. A simple mechanical structure was designed in AutoDesk Fusion 360. The assembly consist of two main parts. A rectangular cavity on top to keep the LCD Screen in place and a magnetic eye tracker holder with limited movement allowance. Second, a housing to place a processing unit and other sensors like vibration

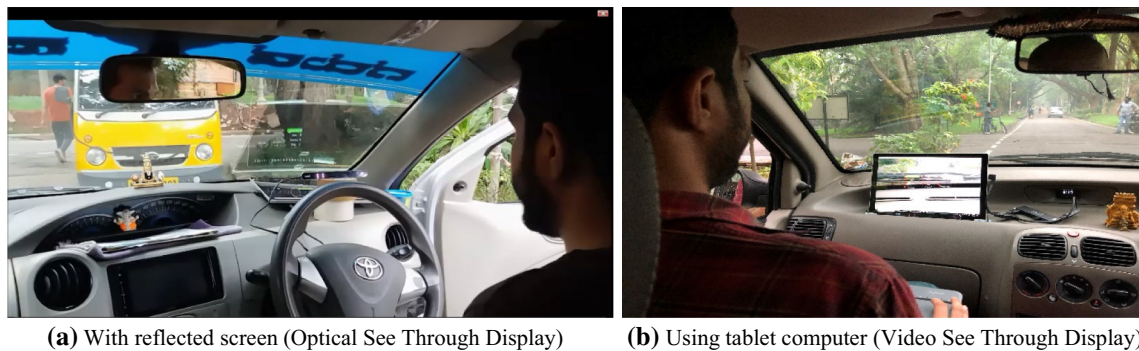


Fig. 1 System setup

or light measuring ones just below the screen. The LCD Screen is reflected in the windshield which can be operated upon by the operator using Eye Gaze Tracking. The assembly comes with an adhesive base to keep the HUD in fixed and stable position. The whole assembly is approximately $31\text{ cm} \times 20\text{ cm} \times 10\text{ cm}$ in size. The eye tracker holder allows the eye tracker to be adjusted according to operator position and convenience. Figure 2 shown the design with all measurements, in particular

- Screen holder is $30\text{ cm} \times 21\text{ cm} \times 3\text{ cm}$ (Rectangular Cavity).
- Eye tracker arrangement is $9\text{ cm} \times 9\text{ cm} \times 3\text{ cm}$ (Sliding assembly).
- Sensor housing at the base is $18\text{ cm} \times 20\text{ cm} \times 5\text{ cm}$.

The base can be modified to place over the dashboard. It can be stuck with strong adhesive or screwed to the dashboard. The screen was used for the simulator while for a real car, only the base and eye tracker holder was used as the display was reflected on the windscreen.

After designing the holder, we tried to find the best position of placing an eye tracker in front of the driver. We did not want to obstruct view of the driver while existing eye trackers cannot track eyes if it is placed more than 100 cm

away from the user. Position of the steering wheel also added challenge in terms of placing the eye tracker.

We set up a CAD model for a sedan car and used a Digital Human Model (DHM, Fig. 3a) with 95% composition male mannequin, set up the seat in at the extreme backward position and generated a track box indicating the visual field as per the specifications defined for COTS trackers (Tobii EyeX, 4c and PCEye Mini). The best position was found to be behind the steering wheel and 910 mm away from male mannequin.

During the study, we used a Toyota Etios car. We checked whether the LCD screen can be powered by the car engine and it can be powered using a standard micro-USB mobile phone charger. We checked the eye tracker position with the highest and lowest positions of the steering wheel and the tracker could track eyes in both positions (Fig. 3b, c). The LCD screen was reflected on the windscreen and we calibrated the eye gaze tracker using our neural network based algorithm for each participant.

4.2 Wireless switch

A wireless switch module was designed for performing selection task while undertaking pointing tasks using eye gaze controlled interface. This switch consists of a NodeMCU v1, a limit switch or touch sensor, SPDT switch

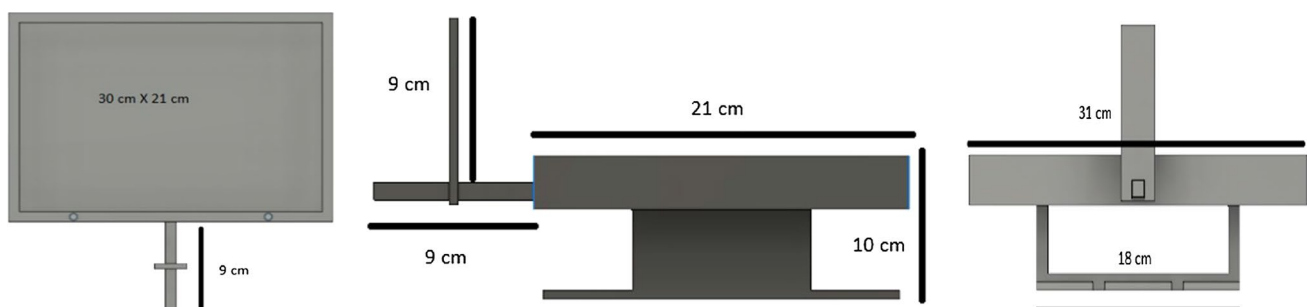
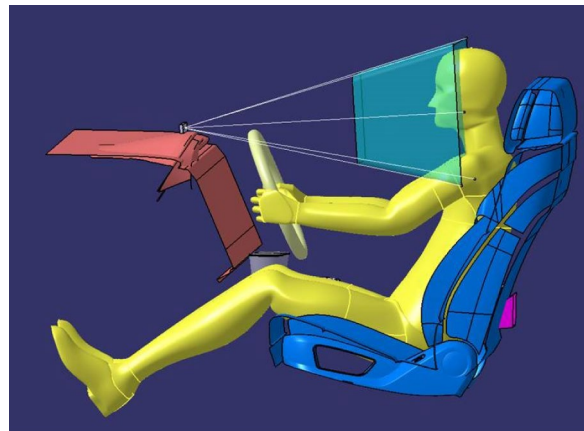


Fig. 2 Hardware part of the invention



(a) Digital Human Model



(b) Lowest position of steering wheel



(c) Highest position of steering wheel

Fig. 3 Measurements of eye tracker position for driver

and a Lipo 3.7v 500mAh battery. The schematic of the module is illustrated in Fig. 4. NodeMCU uses WiFi to communicate. The touch sensor or limit switch which is used to initiate the trigger is connected to NodeMCU. The signal pin, Vcc and ground of touch sensor/limit switch are connected to NodeMCU respectively. The positive and negative terminals of battery is connected to Vin and ground pins of NodeMCU via a SPDT switch. The NodeMCU is programmed to act as an access point and a server with the name “Wireless Switch”. The client connecting to the access point will receive “0” until it requests root “/” address of the server. As soon as the switch is triggered from the server, it sends “1” until the client acknowledges by requesting “/rst” address of the server after which it sends “0”. From the client, when it gets “1”, we perform the left mouse button down followed by up i.e., left click of a mouse. So a left mouse click is made wherever the cursor is pointing on the screen as soon as we press the trigger switch from the wireless switch module. The client program is written in python using HTTP and an exe file is built. Two prototypes are shown in Fig. 5, the actuation can be done either using a limit switch or a touch sensor.

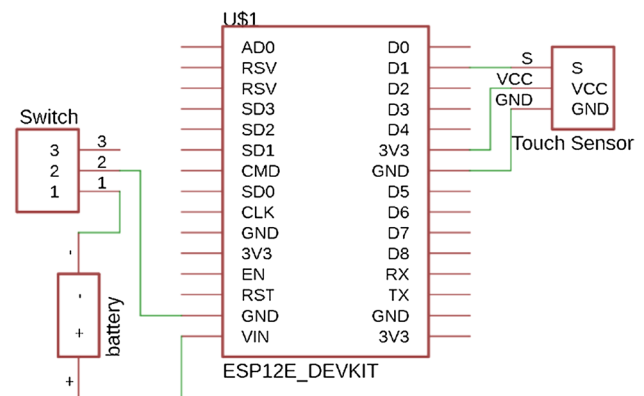


Fig. 4 Schematic of wireless switch

4.3 Multimodal software

The software was selected based on our prior research on gaze controlled interface and different multimodal fusion strategies [6]. We used commercial of the shelf (COTS) sensors to track finger and eye gaze. If the user puts his hand on the finger tracking sensor, the pointer stops moving based on

eye gaze. When the user removed his hand from top of the finger tracking sensor, the pointer resumed to move based on eye gaze of user (Fig. 6).

4.4 Eye gaze tracking system

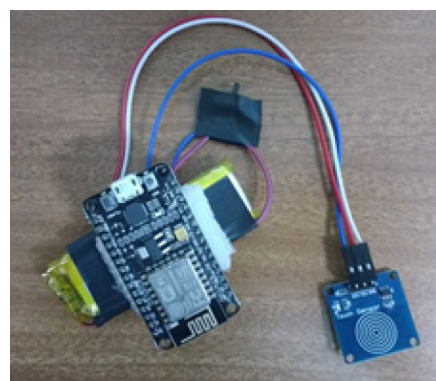
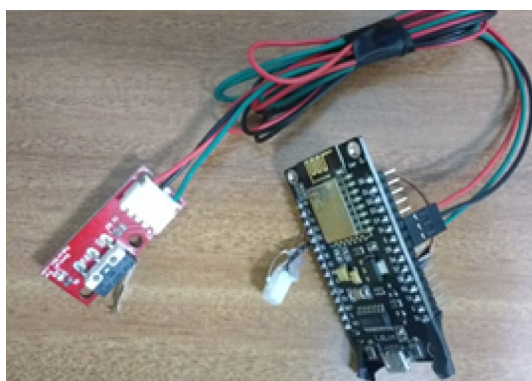
Screen mounted eye gaze trackers are intended to be attached below a flat computer screen. However, we intended to control cursor on a windscreen, which will not be flat like a computer monitor. We also could not attach the eye tracker on the windscreen itself as it will be too far away from the driver. Hence we developed our own calibration system to track eye gaze on a curved surface.

We developed a socket program using the API of the eye tracking vendor and transform the output of the raw gaze coordinates based on screen resolution and zooming level of the rendering device. Example of transformation includes multiplying normalized coordinates by fixed constant and add fixed offset if the screen is inverted for mirroring on HUD. Next, we developed a 9-point calibration application that displayed nine squares one at a time on screen. Each square was of size 50×50 pixels and displayed for a duration of 2.5 s. Users were instructed to fixate attention on the square. Users eye gaze was recorded for each square and the median gaze position was calculated for last 2 s of a square's displayed duration. We set up input files for our models by recording the mid points of each square and the median gaze coordinates recorded by the gaze tracker transmitted by the socket program. During model development,

we used a desktop computing environment where both wearable and screen mounted trackers can operate. We recorded data from 8 users using a screen mounted (Tobii PC EyeX Mini [36]), accuracy 0.3° of visual angle, frequency 30 Hz) eye gaze tracker. We set up four combinations of input and output as follows:

1. Predicting x-coordinate in screen from x coordinate recorded by gaze tracker
2. Predicting x-coordinate in screen from x and y coordinates recorded by gaze tracker
3. Predicting y-coordinate in screen from y coordinate recorded by gaze tracker
4. Predicting y-coordinate in screen from x and y coordinates recorded by gaze tracker

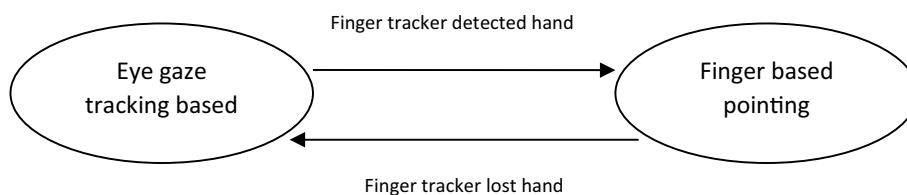
For each combination, we trained a linear regression and neural network model with 9 data points recorded from the calibration program separately for each participant. We used the fitlm function in MATLAB for creating a linear regression model. We set up an artificial neural network with a single hidden layer with five neurons using Google Tensor Flow. The learning rate was set at 0.01 and the model was trained for 1000 epochs. The Gradient Descent Optimizer from TensorFlow API is used for this purpose. We tested R2 and RMS error values for predicted x and y coordinates for the same 9 points used for training the model. Finally, we calculated the average R2 and RMS values considering all participants. Figure 7 below plots these values for each case.



(a) Prototype of wireless switch module with limit switch (b) Prototype of wireless switch module with Touch sensor

Fig. 5 Prototype of the wireless switch

Fig. 6 State chart diagram of eye gaze–finger tracking combined system



It may be noted that the neural network model taking x and y coordinates as input produced higher value of R^2 and lower values for RMS error than the linear regression model in each case. We implemented a neural network model in C# and compared its output with the Tensor Flow models. The cursor control system is implemented in C# and initially runs the 9-points calibration program and uses this to train the neural network. It also records offset of the predicted values for each 9 points. Then it checks the accuracy of the prediction and returns to the calibration routine if the R^2 value of prediction is less than 0.9 and RMS error is greater than 100. After successful training of the network, it records gaze coordinates in real time from the gaze tracker and runs the network. For each predicted value, it corrects the offset by comparing the predicted value and the offset from its nearest calibration point. Finally, the system activates the target nearest to the present cursor position. When a target is activated, the target can be selected by pressing a wireless switch attached to the steering wheel. The algorithm is explained in the flow chart in Fig. 8 below.

4.5 Finger tracking software

We used the following set of equations to take an orthogonal projection of 3-dimensional finger position measured using a Finger tracking controller on a 2-dimensional screen.

$$ScreenX = \frac{ScreenWidth}{w} \times (finger.TipPosition.x + a)$$

$$ScreenY = \frac{ScreenHeight}{h} \times (b + c \times finger.TipPosition.y - d \times finger.TipPosition.z)$$

The constants a , b , c , d , w and h were calculated based on the relative screen position with respect to the finger tracking sensor.

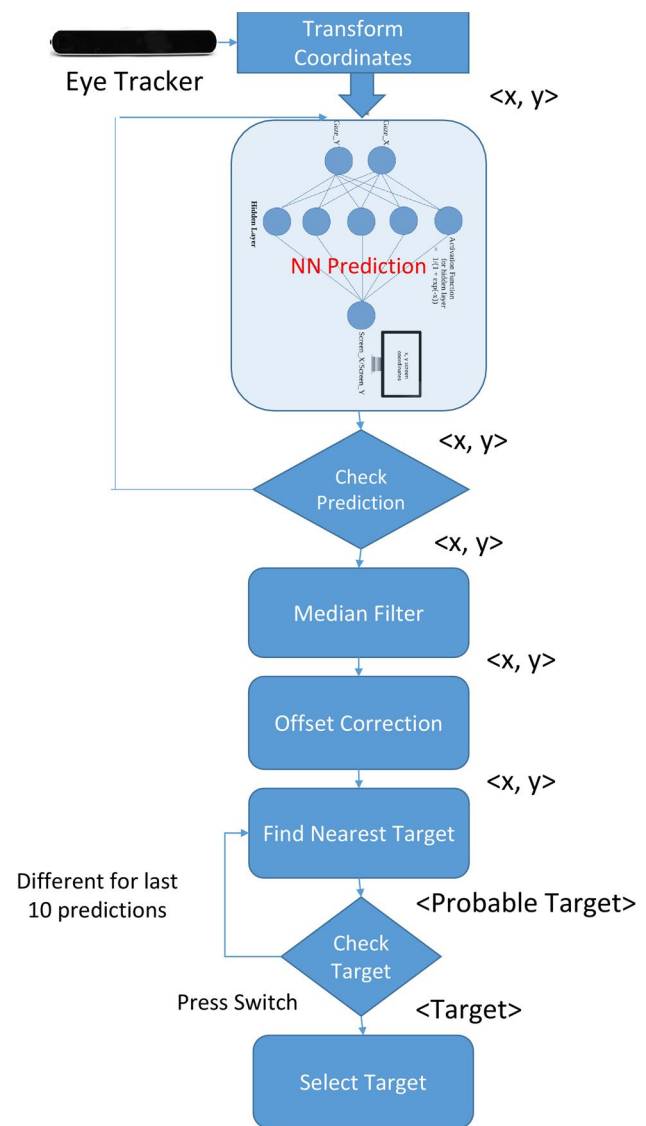
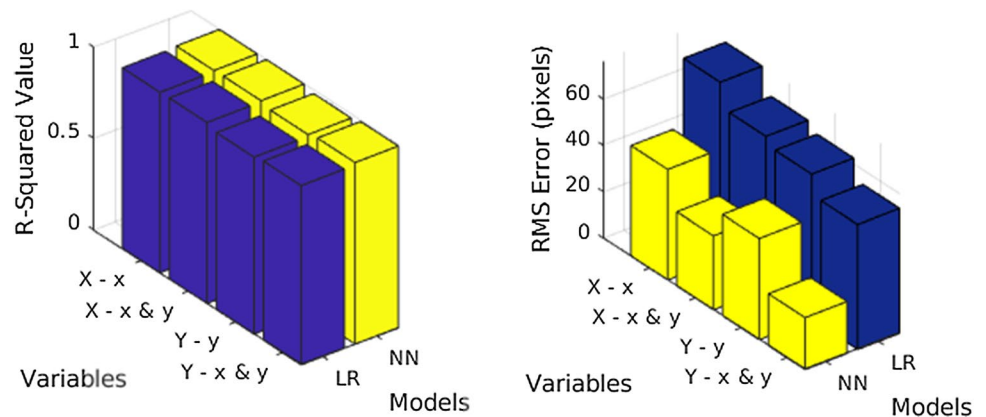


Fig. 8 Controlling cursor in a curved plane

Fig. 7 Comparing models to develop cursor control device



4.6 Improving eye gaze controlled interface

In user studies involving gaze controlled interface, we noted that as users stare at the middle of the target, due to the inaccuracy of the tracker or their head movement, the neighbouring button was occasionally selected. This probability of wrong selection increases if the buttons are closely spaced in the interface. Hence, the probability of wrong selection will be reduced if we can increase the inter-button spacing. However, we cannot change the design of an interface just to make a particular interaction technique work better.

We have explored the option of introducing hotspots inside each button to facilitate eye gaze tracking interaction. If we can introduce a hotspot on each button and keep them well separated, we can instruct users such that the first saccade on a button would launch on these hotspots. We hypothesize that keeping these hotspots well separated may reduce chances of wrong selection. The hotspots will not change the size of any screen element or the interface, rather aim to leverage the ‘pop-out’ effect of parallel visual search. Hence, they will not require larger length of saccades compared to the same interface without hotspots.

To find the best position of hotspots, we have represented an interface as a graph where each node corresponds to a target button (clickable objects) and neighbouring buttons are connected by an edge. For example, the interface in Fig. 9 corresponds to the following graph.

We assumed each button has a hotspot on it, which is initially located at the centre of the button. The weight of

each edge is equal to the Euclidian distance between the hotspots of two neighbouring buttons. We explored two different algorithms to increase the distances between hotspots. We have defined the following cost function and tried to minimize it.

Cost Function = $\sum_{i,j} d_{ij} \frac{1}{d_{ij}^2}$, where d_{ij} is the distance of the hotspots between buttons i and j and is equal to the weight of the edge between nodes i and j .

We have modelled the problem of finding optimum locations of hotspots as a state space search problem. Each state corresponds to a particular organization of hotspots. A state transition occurs when any hotspot changes its position. If we consider that each button has k possible positions and if an interface has n buttons, then an exhaustive search algorithm needs to evaluate kn states. Even for a moderately complex interface, an exhaustive search algorithm will be computationally intensive or almost impossible. Hence, we used the following two algorithms.

Greedy algorithm This algorithm picks up the edge with minimum weight, which means the two most closely spaced buttons. It checks the degrees of the two nodes of the minimum-weight edge and updates the hotspot of the node with higher degree. The algorithm calculates the centroid of the hotspots of neighbouring nodes of the selected node and the new hotspot is calculated as the nearest point on the selected button (or node) to the centroid. While selecting the next node for updating the

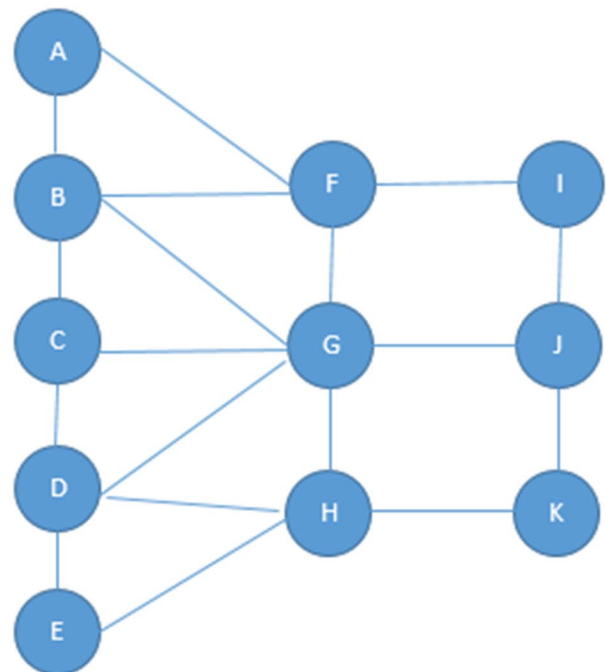


Fig. 9 Graph structure of an interface

hotspot, the algorithm checks whether the node is visited earlier and, if so, it selects a different node. The algorithm is greedy in the sense that it only updates the hotspot if the overall value of the cost function is reduced from the previous value.

Simulated annealing This algorithm randomly selects a node and also randomly selects a point on the node as its new hotspot. If the new hotspot reduces the value of the cost function, then it is selected and updated. However, even if the new hotspot increases the value of the cost function, it may still be selected based on the following condition:

$$e^{\frac{(\text{oldCostFn} - \text{newCostFn})}{T}} > \text{A random number between 0 and 1.}$$

In the above equation, the value of T runs from 5000 to 1 and reduced by 1 in each iteration.

Both algorithms were run for 5000 iterations and Fig. 10 below plots the values of the cost function for the minimum value of the cost function obtained in the algorithms. In this figure, ‘Original’ stands for the initial state of the algorithm when all the hotspots were in the middle of each button.

We also plotted the sum of the weight of edges of the graph for the minimum value of the cost function for each algorithm in Fig. 11 below.

It may be noted that both algorithms reduced the cost function and increased the overall weight of edges. The Simulated Annealing (SA) algorithm reduced the cost function further than the greedy algorithm. The greedy algorithm was stuck in cycle and a local optimum after visiting all nodes a couple of times. The SA algorithm never stuck in the local optimum due to randomly choosing node and hotspots, although we also could not conclude whether it reached the global optimum. It may be noted that the weights of edge in the final state of SA are significantly lower ($p < 0.01$) from the initial state in a paired t test. Figure 12 below shows final positions of hotspots on the buttons as red dots.

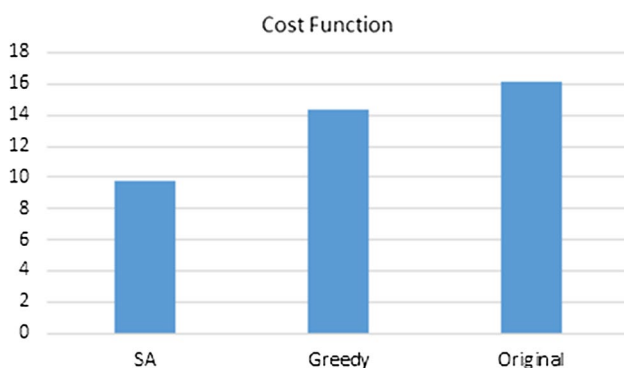


Fig. 10 Comparing the cost function for different algorithms

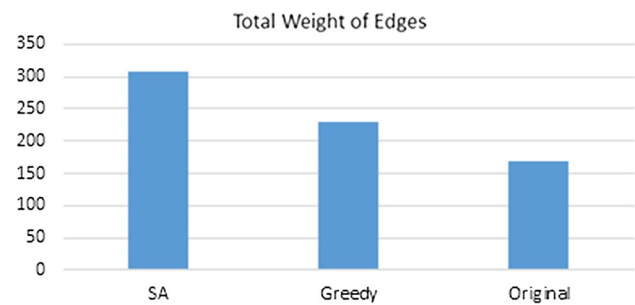


Fig. 11 Comparing total distances among hotspots for different algorithms

5 Content design for interactive HUD

5.1 Designing icons

The content design of the interactive HUD should not only be perceived easily by the user but should also be visible in all lighting conditions. It should not obstruct the driver's view. The interface is designed to comply with the above requisites and to bring a common design for both modes of interface rendition viz.

- (i) The reflected interface and
- (ii) The tablet interface with back camera.

A survey was conducted through an online system for persons with driving experience and data was collected in terms of user interaction requirements and functions required to be displayed in the dashboard as pointed below:

1. Easy to recognize icons from their prior learning experiences which they can relate to.
2. Icons should be compatible in the car environment among various other existing icons.
3. Avoiding any overlap in the icons with other functions which can call for confusions.

The survey collected data from 90 participants. Figure 13 represents the data collected from users in terms of the required functionalities on the dashboard. Before starting and after ending the trip users prefer to operate seat adjustment, window glass rolling and AC vent control options, checking proximity sensor alert and tyre pressure status. While driving users prefer to have dashboard controls for navigation, music, phone notifications, distraction indication, speed indication, fuel indication, window glass rolling options, battery indication, vent control options, warnings and alerts to avoid accidents, headlight adjustments, glass defrosting options and reminder alerts.

Fig. 12 Final positions of hot-spots after simulated annealing

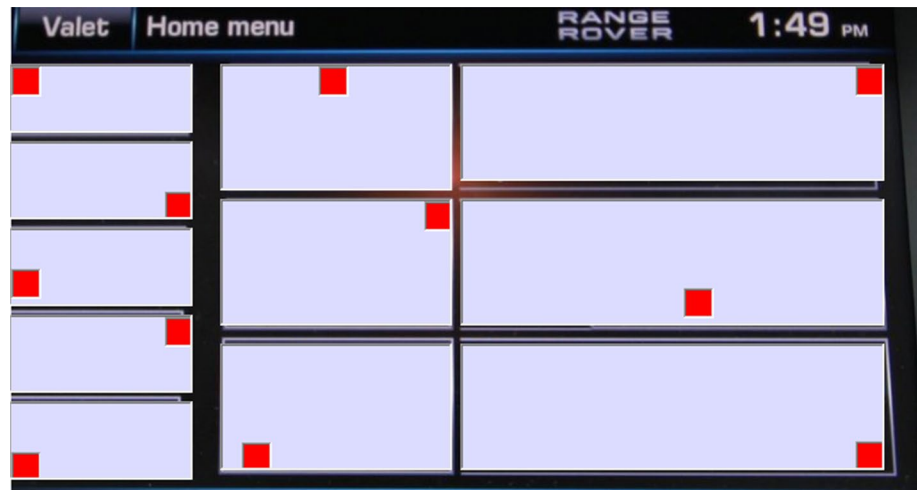
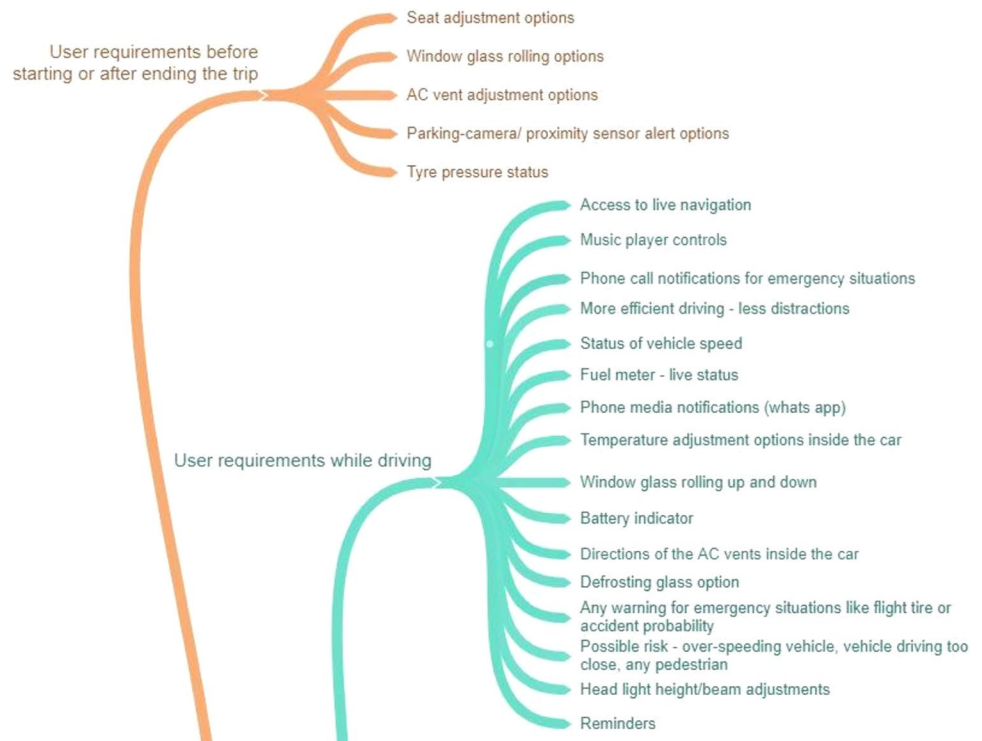


Fig. 13 User requirements



We selected a set of functionalities to be offered from the HUD and designed four different Glyph icons [9] for each function. We undertook another survey on icon designs and asked participants to select the most appropriate and relatable one for each function. Figure 14 shows a snippet from the survey and Fig. 15 shows a graphical representation of users' response. The most popular icon for each function is incorporated into the HUD.

5.2 Adaptation for different ambient lighting conditions

The difference in lighting conditions on the interface can originate from a variety of sources in the driving environment and can hinder the driver's visibility of the external environment. It can also induce discomfort and can make the driver weary. This section elaborated on ways to

Questionnaire

Fig. 14 Questionnaire snippet

compromise both the reduced legibility and conspicuity of display information due to glare on the display and the reduced visibility of the environment and increased physical discomfort caused by glare from car head up displays. The available research [1, 29] on this topic suggests that this can be achieved through the following strategies for alleviating glare on displays due to ambient lighting conditions:

1. Providing adequate display luminance.
2. Using high contrast display technologies to ensure adequate contrast.
3. Placing or project displays to a position which can minimize exposure to glare.
4. Using anti-glare coatings or films to filter incoming light and reduce glaring reflections from the display.
5. Displaying the content using a dark background to minimize the luminance emanating from the display.

In order to mitigate the issues caused by varying lighting conditions, a dynamic contrast display which would adjust automatically to the lighting conditions on the interface was designed. The display would check constantly for the lighting conditions at the surrounding points of an icon and will transform the icon colour to a reverse contrast colour. The icons change from dark to light and vice versa according to the lighting conditions. The brightness of each icon was calculated using the Brightness method in Emgu.CV library. This method gets the hue-saturation-lightness (HSL) value for the corresponding colour structure. Any value above the normal lighting condition of 0.7 renders the icon as dark and anything below this value restores the white colour of the icon. Figure 16 shows the reverse contrast display of icons with respect to the surrounding lighting conditions.

5.3 Icon size

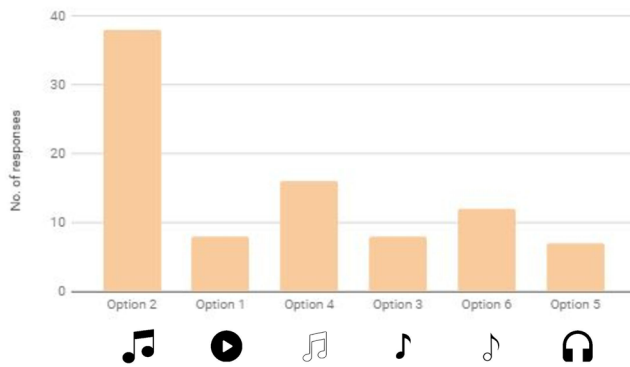
Legibility and conspicuity of the icons is vital for visibility and perception. Legibility is not confined to visibility or detection; it is also the capability of being able to discern shapes or characters based on appearance. Both graphical symbols and text which makes up an icon are vital sources in influencing the legibility of the icon. The calculations are made with reference to existing NHTSA guidelines on Human Factors Design Guidance for Level 2 and Level 3 Automated Driving Concepts [11]. The research on this topic suggests that this design goal can be achieved by having icon sizes with optimal visual angle of primary graphical elements at 86 arcmin. This can vary for time-critical applications and non-time-critical applications. For time-critical applications, the minimum visual angle for primary graphical elements is calculated to be 41 arcmin, while for non-time-critical applications, it is 34 arcmin. As discussed above, the size of the text is also important for icon legibility. While the optimal height of text is calculated as 20 arcmin, it is 16 arcmin and 12 arcmin for time-critical and non-time-critical applications respectively. Table 1 shows the values of these parameters.

Table 2 shows the different visual angles for different applications. Calculations were made using optimum angle as it gives more legibility and conspicuity. Table 3 shows the calculations involved in determining the icon size. The calculations take into consideration the specifications of the tablet used for display. The resolution of the display used is 2736×1824 pixels with a DPI (Dots Per Inch) of 267 and a scaling factor of 200%. Table 4 shows the measured and calculated values of the icons. It can be seen from the table that the real value of the displays concurs to the calculated values. A deviation from the calculated values can reduce the legibility and thereby cause discomfort to the driver.

5.4 The head up display interface

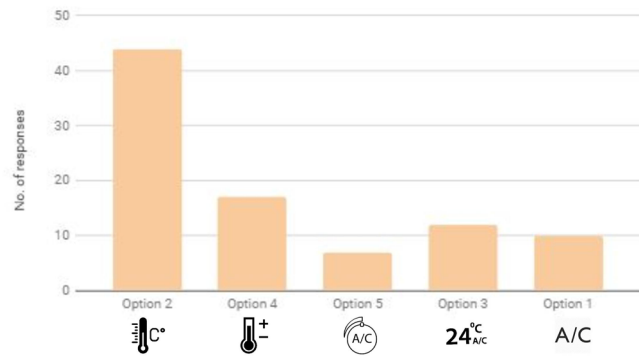
Figure 17 shows homepage of the final UI for the HUD by encompassing all the above calculations and considerations for icon symbol, icon size and ambient lighting conditions. The icons in the UI are selected based on the results obtained from the survey. All icons are placed on the screen in an elliptical fashion such that it covers the entire screen and are spread out enough to provide easy selection feedback while moving using eye gaze. It can be seen from the picture (Fig. 17) that the icons with bright light conditions have dynamically set to the reverse contrast condition (dark icons) and vice versa. The red dots on the icons are placed based on the state space search algorithm discussed in Sect. 4.6 of the paper. The UI has icons for vent control, climate control, seat control, settings, maps, music/video, deco lights, emergency vehicle and an eye icon for indicating distraction.

Q 1. Choose the most appropriate icon for music



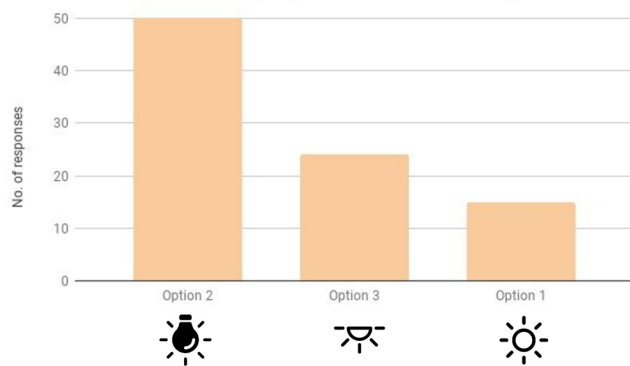
(a) Results for music icon

Q 2. Choose the most appropriate icon for climate control



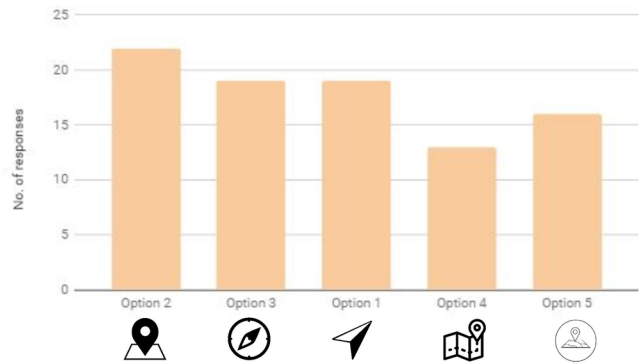
(b) Results for climate control icon

Q 6. Choose the most appropriate icon for deco. light



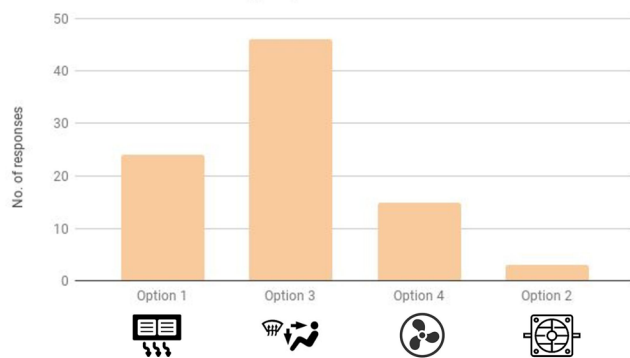
(c) Results for decoration light icon

Q 3. Choose the most appropriate icon for maps



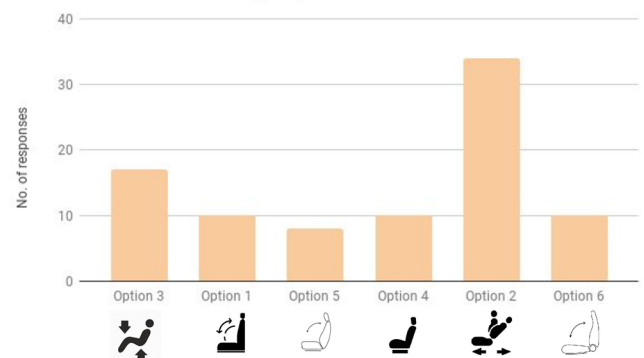
(d) Results for navigation icon

Q 4. Choose the most appropriate icon for vent control



(e) Results for vent control icon

Q 5. Choose the most appropriate icon for seat control



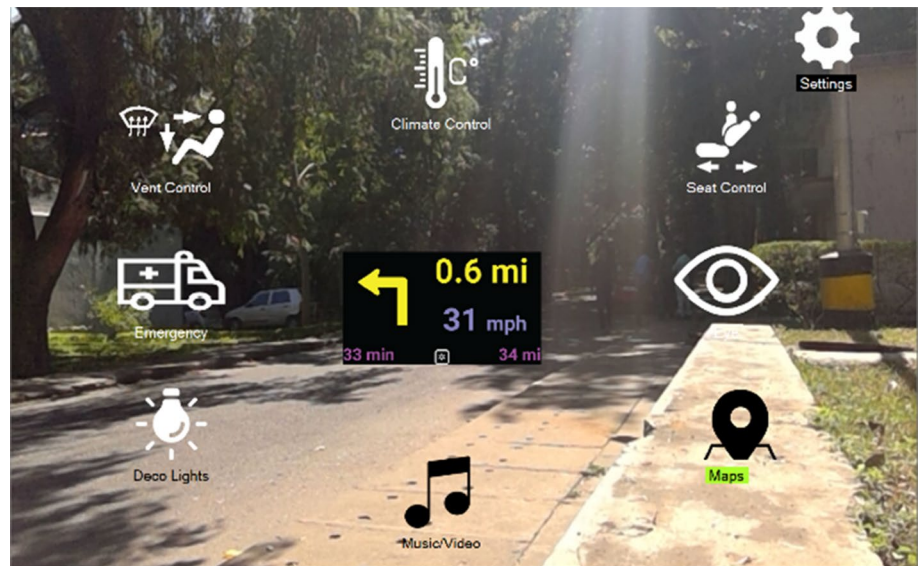
(f) Results for seat control icon

Fig. 15 Graphical representation of the results collected

The progress bar in the picture shows an estimation of the cognitive load of the driver. Figure 17 shows the Climate Control icon and Maps icon in the dark/black colour against the bright background. All the other icons are set to white colour due to dark/ill-lit background condition.

The eye icon will blink in red colour if the driver's gaze is off the road for more than 2 s or if the driver has fallen

asleep. This acts as a measurement for driver's distraction. An alert sound also goes off during this process. It falls back to normal once the driver's attention is back on the road. The emergency vehicle icon would blink with directions when an ambulance is passing nearby. The direction indicator which would appear with the icon will indicate the direction from which the ambulance is coming using sound detectors placed

Fig. 16 Reverse contrast icons**Table 1** Summary of user studies

Study no.	Position of participant	Environment	Modalities of interaction	Aim
1	Driver	Simulator	Eye gaze tracker, finger tracker, touchscreen	Comparing response times and driving performance of new system with existing touchscreen system
2	Passenger	Inside vehicle	Eye gaze tracker, touchscreen	Investigating effect of in-car vibration for eye gaze tracking system
3	Passenger	Inside vehicle	Finger tracker	Investigating effect of in-car vibration for finger tracking system
4	Driver	Inside vehicle	Eye gaze tracker	Finding best position of eye gaze tracker for driver and measuring its performance for the head up display
5	Driver	Simulator	Eye gaze tracker, finger tracker, touchscreen	Comparing physiological parameters estimating cognitive load of driver between proposed system and existing touchscreen system

Table 2 Visual angle settings for icon and text size

Parameter	Icon	Text
Optimal visual angle	86 arcmin	20 arcmin
Minimum visual angle for time critical applications	41 arcmin	16 arcmin
Minimum visual angle for non-time critical applications	34 arcmin	12 arcmin

outside the car. A demonstration video of the whole set up can be found at http://cambum.net/IHUD_May19.mp4.

6 User studies

We conducted five user studies as follows

Table 3 Calculations for icon and text size

Parameter	Icon	Text
Optimal visual angle in degrees (x)	1.433	0.333
Tan(x)	0.025	0.0058
Symbol height (H)	$D \times \tan(x) = 120 \times 0.025 = 3 \text{ cm} = 0.275 \text{ in.}$	$D \times \tan(x) = 120 \times 0.0058 = 0.7 \text{ cm} = 0.275 \text{ in.}$
Symbol height (H) in pixels (dpi for display = 267)	320.4	72.09

Table 4 Calculated size of icons for HUD

	Calculated value	Real value on surface pro display	Real value on reflective display
Icon size			
cm	3	4	$1.8 \times 2 = 3.6$
Inches	1.2	1.57	1.417
Pixel	320.4	419.19	378.339
Text size			
cm	0.7	0.8	$0.35 \times 2 = 0.7$
Inches	0.275	0.393	0.275
Pixel	73.425	104.9	73.425

1. Study involving a simulator comparing the interactive HUD with and without hotspots and touchscreen system for a existing JLR dashboard
2. Study inside a running vehicle with the eye gaze tracking system set up for passengers
3. Study inside a running vehicle with the finger tracking system set up for passengers
4. Study inside a static vehicle with the system set up for the driver
5. Study involving a simulator comparing physiological parameters estimating cognitive load of drivers for the final prototype and an existing touchscreen system

The first study compared the new system in a dual task setting. The second study evaluated the system inside the car and with respect to on-board vibration. The second and third studies separately evaluated the eye gaze and finger tracking system inside the running vehicle. The fourth study evaluated the performance of the multimodal

system by positioning the sensors and display with respect to a driver. In the following sections, we have described these studies.

6.1 Study 1: driving simulator study with dual task

Participants We collected data from 11 participants (8 male, 3 female, average age 29.2 years). All participants used to drive cars and had no problems to use the experimental set up.

Material We used a Tobii EyeX eye gaze tracker and Lenovo Yoga Laptop for the secondary task. A Logitech G4 driving wheel and associated pedals were used for the primary driving task. We used the ISO 26022 lane changing task to measure participants driving performance and it was run on a 40" MicroMax TV screen (Fig. 18). Selection was done by pressing a switch on steering wheel as a previous study [5] found hardware switch based selection is faster compared to dwell time or voice based selection for gaze controlled interface.

Design In this dual task study, participants undertook ISO 26022 lane changing task as their primary task. In this task participants needed to drive along a 3-lane motorway. At a regular interval, drivers were shown a sign and instructed to change lane. The destination lane was randomly chosen and the driving path is automatically logged.

The secondary task involved pointing and clicking on a screen. We allowed users to use either eye gaze or finger tracker. We used the following three modalities for pointing

Fig. 17 Final UI with dynamic reverse contrast display

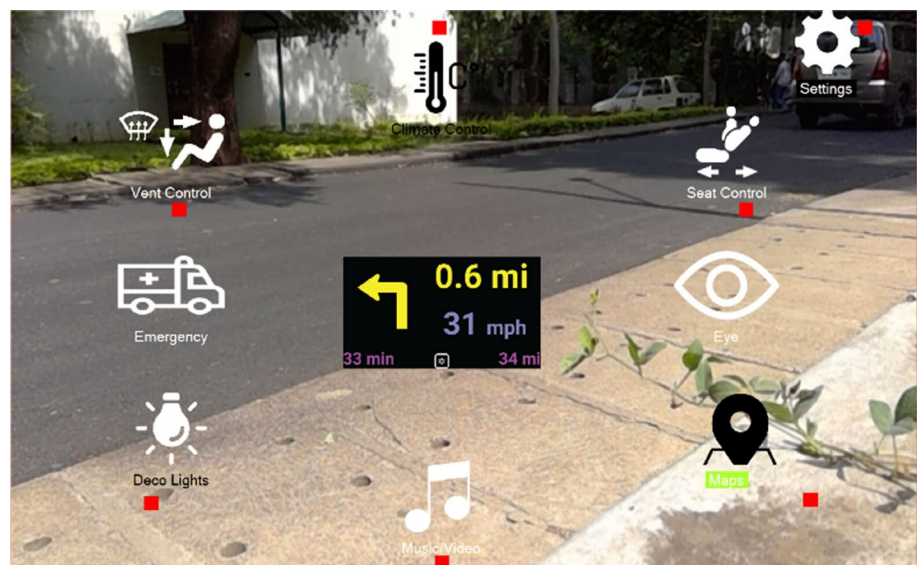




Fig. 18 Set up of gaze controlled projects display

1. Eye gaze and finger tracking with hot spots
2. Eye gaze and finger tracking without hot spots
3. Touching

We have used an existing dashboard display from Jaguar Land Rover (Fig. 12). We used the same dimensions of the buttons as the original display but removed all button captions. During the study, one of the buttons of the display was randomly selected as target and a caption ‘Target’ appeared on that button. The primary and secondary tasks were linked through an auditory cue. While driving, participants were instructed to point and select the designated target on the secondary screen after hearing an auditory cue. The auditory cue was set to appear between 5 and 7 s interval. The auditory cue was kept on repeating during the whole duration of driving.

Procedure Initially we explained the aim of the study to participants. They were first trained with the gaze controlled interface and allowed to use the driving simulator to undertake a test run. After training, they undertook trials in different conditions in random order. Participants were instructed to drive realistically without veering off from the driving lane. While using the gaze controlled interface with hot spots, participants were instructed to focus on the hot spot of the designated target while no such instructions was given in other two conditions. After each condition, participants filled up the NASA TLX and SUS questionnaire [35].

Results We have measured the following dependent variables

1. *Driving performance* is measured as
 - (a) *Mean deviation* from designated lane calculated according to Annex E of ISO 26022 standard.
 - (b) *Average speed* of driving, we investigated if the new modality significantly affected driving speed.
 - (c) *Standard Deviation of Steering Angle*, a large standard deviation means drivers made sharp turns for changing lanes.

2. *Pointing and Clicking performance* is measured as
 - (a) *Error in secondary task* as the number of wrong buttons selected compared to total number of selections.
 - (b) *Response time* as the time difference between the auditory cue and the time instant of the selection of the target button. This time duration adds up time to react to auditory cue, switch from primary to secondary task and the pointing and selection time in the secondary task.
3. *Cognitive load* measured as the NASA Task Load Index (TLX) score.
4. *Subjective preference* as measured as the System Usability Score (SUS).

For each dependent variable, we initially compared the descriptive statistics and then undertook parametric or non-parametric statistical hypothesis test. If ANOVA reported a significant difference, we used t test for pairwise comparisons.

Before undertaking the trial in dual task condition, participants used the driving simulator for undertaking the primary driving task only. The driving path of this single task situation was used as a baseline for evaluating degradation in driving performance for the secondary task. Following the description of ISO 26022 Annex E, a reference path trajectory with a constant lane change start position and lane change length was calculated, which has a maximum correlation with the base line driving path. For example, in Fig. 19, the green line shows the reference path while the red line is the driving path in dual task situation.

The driving performance (Fig. 20) in terms of mean deviation from designated lane was significantly different in a Kruskal–Wallis H-test [$\chi^2(2,28) = 10.56, p < 0.05$]. Pairwise signed rank test also found that driving performance was significantly different between projected gaze controlled systems for using hotspots. It may be noted that

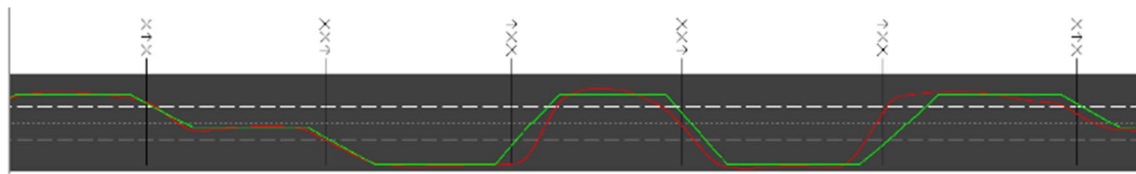


Fig. 19 Illustrating calculation of mean deviation from lane to measure driving performance

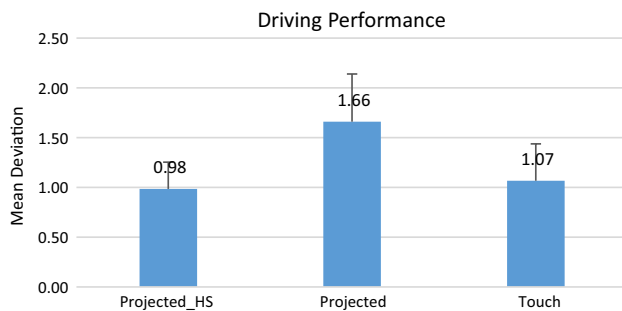


Fig. 20 Comparing driving performance in terms of mean deviation

using hotspots, mean deviation from designated driving lane was reduced by 41% for projected gaze controlled interface and the mean deviation for hotspot equipped projected gaze controlled interface was even lower than the touchscreen based system.

The differences among average speeds of driving were not significant. It was approximately 3 kn/h slower for hotspot equipped projected gaze controlled system than touchscreen system.

The standard deviation of steering angle was significantly lower for projected gaze controlled system than the touchscreen system in a signed rank test ($p < 0.05$). It may indicate drivers needed to drive more carefully with the projected gaze controlled system compared to touchscreen system.

The average number of wrong selections was higher in projected gaze controlled interface than touchscreen system although using hotspots reduced it by 15.6% than without it.

Average response time was still lowest for touchscreen based system but it is only 2% higher in hotspot equipped projected screen (Fig. 21). A single way ANOVA found significant difference among the reaction times [$F(2,257) = 4.84, p < 0.05$]. A set of unequal variance t tests found that touchscreen had significantly lower response times [$p < 0.05$] than projected screen without hotspot while the difference between touchscreen and hotspot equipped projected screen was not significant.

The TLX scores (Fig. 22) were highest for projected gaze controlled system and lowest for touchscreen. The hotspots reduced the average cognitive load by approximately 6% from the projected gaze controlled system without hotspots. However, we did not find any significant difference among

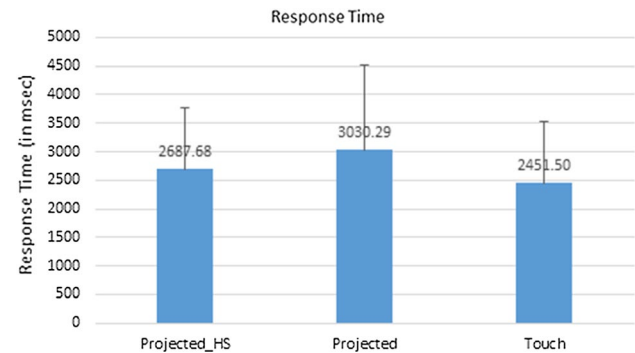


Fig. 21 Comparing average response times

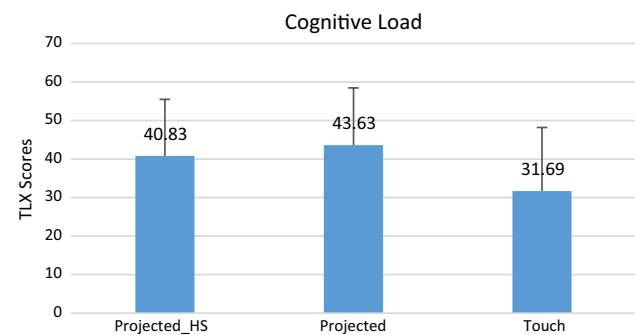


Fig. 22 Comparing cognitive load in terms of average TLX scores

different components of TLX scores for hotspot equipped gaze controlled projected system although the temporal demand (mean 52.89, stdev 19.38) was still higher than other components. A separate study described in Sect. 6.5 compared physiological parameters like pupil dilation and EEG to compare cognitive load between touchscreen and interactive HUD system.

The SUS scores (Fig. 23) were greater than 68 for all cases and highest for hotspot equipped gaze-controlled system, which means all systems were usable with hotspot equipped gaze-controlled interfaces was most preferred by users.

Discussion This study demonstrates that eye gaze tracking devices can be used for projected displays as well. While participants used the projected display, they need not to look

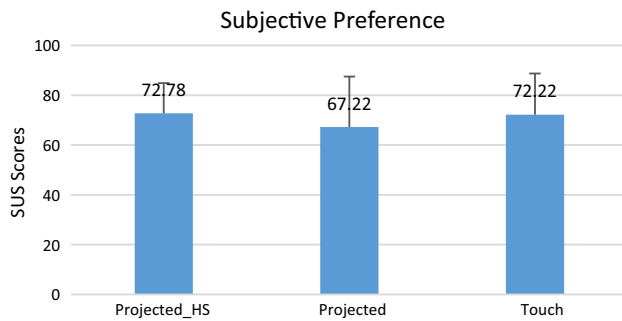


Fig. 23 Comparing preference in terms of average SUS scores

down, stretch hand and could see through the road behind the semi-transparent display. We indeed found that driving performance was better than touchscreen for the hot-spot equipped projected gaze control system. The average response time was only 2% higher than touchscreen system. The SUS scores also indicated users did not face any serious trouble in using this system. The cognitive load was still higher than touchscreens but it should also be noted that our participants use touchscreen enabled device everyday while they were using the eye gaze controlled interface for first time during the trials.

Our algorithm involving hotspots tried to reduce number of wrong selection due to inaccuracy in eye gaze tracking. It leverages the pop-out effect of visual search by placing a set of well separated spots on target buttons of a screen. We explored state space search algorithms to find the best positions of these hot spots on the interface. Our algorithm defined an energy parameter that is inversely proportional to distance between neighbouring hot spots and tried to minimize overall energy by increasing separation among hot-spots. The Metropolis Keyboard [42] used similar concept of minimizing energy in optimizing positions of buttons in a virtual keyboard. Our first trial confirmed that hotspots indeed reduce the number of wrong selections in a gaze controlled interface. We also tried to project the content of dashboard on the windscreen so that drivers need not to look down and reported eye gaze can also be used to control a projected display. Using hotspots with the projected display improved driving performance even compared to touchscreen.

The main problem with the projected display was the number of wrong selections. Using hotspots significantly reduced it but it was still higher than both touchscreen and screen based gaze controlled system. Actually, we could not completely eliminate the skewness of the projected display and positioning the projector without hampering driving equipment was also a challenge, which all resulted in poor calibration of the system. The following studies evaluated the system inside vehicle for both drivers and passengers.

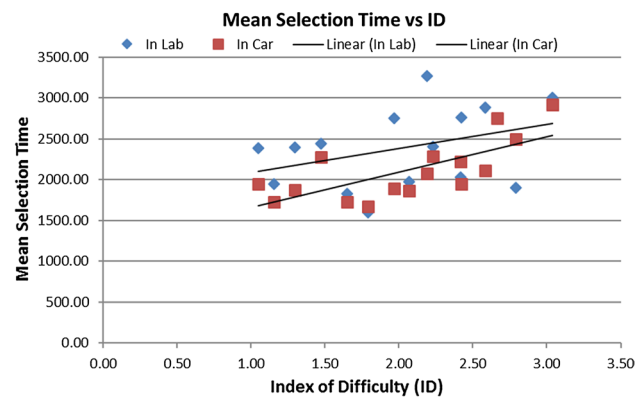


Fig. 24 Comparing selection time of targets using eye tracker inside and outside car

6.2 Study 2: eye tracking system in running vehicle

Participants We collected data from 9 participants (5 male, 4 female, average age 29.8 years).

Material We used a Toyota Innova car and a Lenovo Yoga laptop with 12" screen for the study. The car was plying through a motorway in between 40 and 60 MpH while participants undertook pointing and selection tasks. We used a Tobii PCEyeMini gaze tracker with the Tobii SDK to record eye gaze locations.

Design This study involved participants to undertake ISO 9241 pointing task. The pointer was moved through eye gaze while the selection was done by clicking the left mouse button. We used the following 4 target size and distances

- Target Size: 45,55,65,75 pixels
- Distance from centre of screen: 80,160,240,325 pixels

The indices of difficulty ($ID = \log_2(1 + \frac{Dist}{Width})$) ranges from 1.05 to 3.04 bits/s. The target was coloured white in black background. A nearest neighbourhood algorithm highlighted the target nearest to the cursor position and selection was done using a physical switch. The target was coloured white in black background. We undertook the user trail inside a moving car as well as inside the lab.

Procedure Participants were instructed to bring the pointer near the target box highlighted in white colour and select the box by clicking the left mouse button.

Results We calculated the mean selection time of each target for each participant and plotted the mean selection time vs Index of Difficulty (ID) in Fig. 24. We can infer from the graph that the performance of eye tracker inside the car did

not deviate much from the performance of eye tracker inside the lab. A pairwise t test did not show any significant difference in the performance of eye tracker between the two environments. This evidence is taken for further exploitation of eye trackers in an automotive environment.

6.3 Study 3: finger tracking system in running vehicle

Participants We collected data from 9 participants (5 male, 4 female, average age 29.8 years).

Material We used a Toyota Etios car and a Lenovo Yoga laptop with 12" screen for the study. The car was plying through a motorway at approximately 60 MpH while participants undertook pointing and selection tasks. We used a LeapMotion finger movement tracker with the LeapMotion SDK to record finger movement.

Design The pointing and selection task was initiated through an auditory cue. The study used an existing KIA dashboard (Fig. 25) and participants were instructed to press a button on it after hearing the auditory cue. The auditory cue was set to appear between 5 and 7 s interval. The target button was randomly selected in the car dashboard. The pointing was undertaken through the finger tracking system as described in the previous section and selection was done through dwelling on the target for 500 ms.

Procedure Initially, the participants were briefed about the study and its goal. We calibrated the eye gaze tracker while the car was not moving and demonstrated the pointing and selection tasks to the participants. They undertook the experimental task once to familiarize themselves while the car was not moving although we discarded the data from this familiarization session.

Once the car was driven to the motorway, we started the actual trial. Participants were instructed to take their hands



Fig. 25 Screenshot used in the study

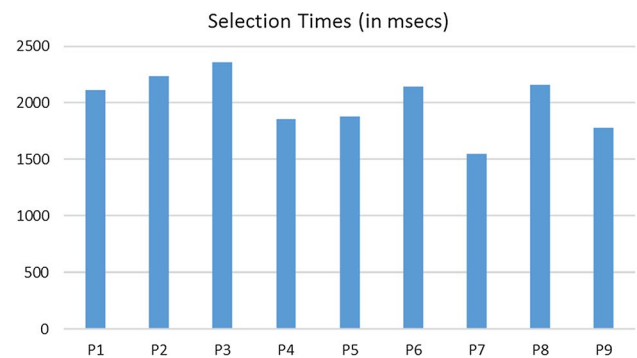


Fig. 26 Response times for individual participant

off from the top of the sensor after each pointing task and start hand movement only after listening the auditory cue.

Results We measured the time difference between the last auditory cue and timestamp recorded at the selection of a target. The average selection times were less than 2 s (Figs. 26, 27).

6.4 Study 4: combined system for drivers in static vehicle

Earlier studies either used a simulator or collected data from passengers. The following study set up the interactive HUD for driver. We collected data from 9 participants seating at the driver's seat in driving position. Among them, there were 6 male and 3 female participants, age range was from 23 to 35 years while height range was from 5'4" to 6'. Finally, we collected data using a standard pointing task displaying 5 targets on screen. Each target was 70 × 70 pixels in size and one of the buttons was rendered differently than other buttons. Participants were instructed first to calibrate using the neural network model discussed in Sect. 4.3 and then to select the designated target as soon as it appears on screen. Participants were also instructed to adjust the seat based on their normal driving posture. We measured differences in

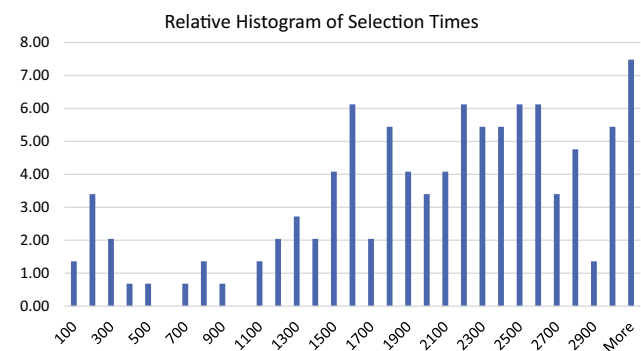


Fig. 27 Histogram of response times

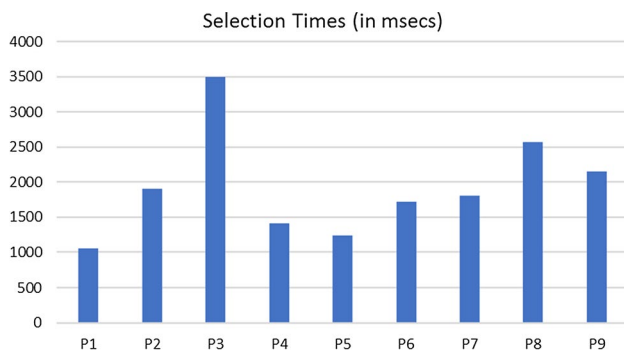


Fig. 28 Response times for individual participant

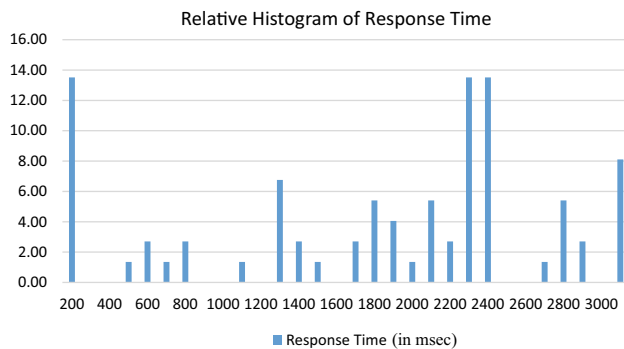


Fig. 29 Histogram of response times

timestamps between appearance of the target and its selection. We collected 81 pointing tasks. Figure 28 below shows the histogram, the median of selection times was 2.1 s, average selection time was 1.8 s and standard deviation was 1.1 s (Fig. 29).

6.5 Study 5: comparing physiological parameters

Study 1 compared the interactive HUD for an existing JLR dashboard and compared response times and driving performance. Studies 2, 3 and 4 undertook studies inside cars and evaluated the response times of the system. This study used a driving simulator similar to the first study but instead of response times it measured physiological parameters

estimating cognitive load of drivers and compared it with a touchscreen system. This study also used the final prototype system and the GUI described in Sect. 5 and Fig. 30.

We used similar protocol and instruments as study 1. We collected data from 11 participants (average Age: 26, 2 females, 9 male) and each participant undertook the lane changing driving task and in parallel undertook five pointing and selection tasks as secondary task. We recorded EEG using an Emotiv Insight dry electrode tracker [14].

We compared the following physiological parameters

- The power spectrum from the Alpha, Low Beta and Theta channel responses from EEG tracker
- The L1 norm of the frequency response of the pupil dilation for both eyes
- The standard deviation of pupil dilation for both eyes

We collected each parameter during the whole duration of the trial and compared their means for all participants. We checked the distribution of data using Kolmogorov–Smirnov test for normality and undertook either sign rank test or two sample t test.

The power of the EEG signal was lower for the Interactive HUD than touchscreen indicating lower cognitive load for all three channels although the difference was not significantly different at $p < 0.05$ in a sign rank test (Fig. 30).

The L1 norm and standard deviation of pupil signal aimed to measure a hike in pupil signal due to increased cognitive load [3, 7, 24]. The values were marginally higher for the interactive HUD than touchscreen although the difference was not significant at $p < 0.05$ in a two sample t test (Fig. 31).

6.6 Discussion

The simulation studies compared the system with respect to a driver undertaking both driving and secondary tasks. Study 1 used an existing dashboard while study 5 used a new user interface developed based on our user survey. We measured response times, physiological parameters and driving performance and compared it with existing touchscreen system. The new interactive HUD neither increased response times,

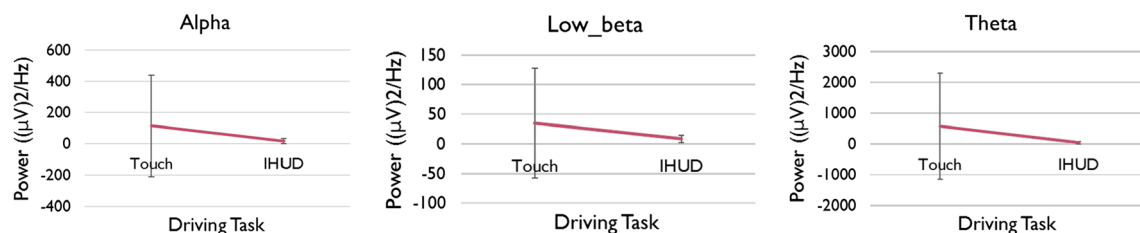


Fig. 30 Comparing EEG response between IHUD and touchscreen systems

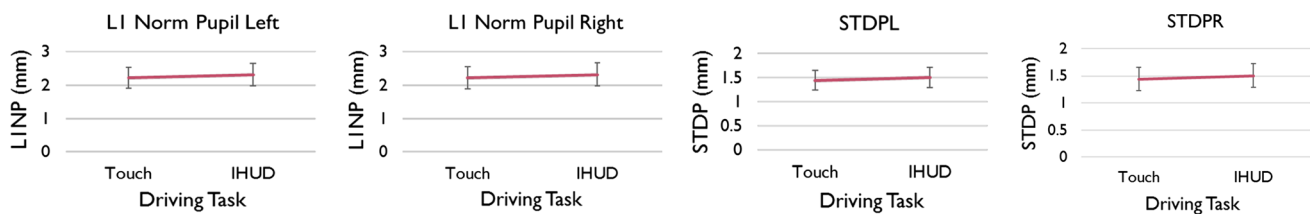


Fig. 31 Comparing pupil dilation between IHUD and touchscreen systems

cognitive load nor degraded driving performance, rather it was improved than touchscreen system.

However, for an actual driving situation, we are still in the process of getting permission for implementing the system for drivers of a running car. Hence, we evaluated the system for passengers of a running vehicle and then for drivers in a non-moving car. We aimed to test performance of the system in automotive environment in particular for the on board vibration and placement of sensors and displays. It may be noted that we used different types of on-screen element for the three studies. The eye gaze controlled system used the standard ISO pointing task, while the finger tracking system used an existing car dashboard. The first study with eye gaze controlled interface found users took longer than 2 s to select icons of small size. So the second study used a realistic display while the required size of the icon was determined according to existing NHTSA guideline as discussed in Sect. 5. We tried to cover different types of existing dashboards as well as proposed a new dashboard from the proposed interactive HUD system. The combined system for drivers was evaluated for only one size of icon. In all studies, the average pointing and selection times were less than 2 s for both drivers and passengers. The pointing and selection times was increased when the target is smaller in size than 70 pixels.

7 Conclusion

This paper proposes a system to facilitate human machine interaction in automotive environment while undertaking secondary tasks. Our proposed system does not require drivers to look down and taking eyes off from road while operating a dashboard. The proposed system consists of a screen whose content is reflected on the windscreen and eye gaze and finger movement trackers to change content of the display based on either eye gaze or finger movement of drivers. The performance of the system was evaluated using a dual task setting involving a driving simulator and it was found that the pointing and selection times are not significantly different between the proposed system and touchscreen display, rather driving performance was improved with the proposed

system than existing touchscreen display. We undertook a series of studies inside vehicle and noted that the performance of the system was not significantly slower for vibrations inside a moving vehicle and in particular, participants can undertake pointing and selection tasks at an average time of less than 2 s inside a Toyota Etios car.

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